Muyu He

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Education

M.S. Computer Science *University of Pennsylvania* Sept 2023 - May 2025

B.A. Philosophy *University of California, Los Angeles* Sept 2018 - Dec 2021

Publications

- **Do Value Vectors in Deep Layers Need Context from the Residual Stream?**
  Muyu He*, Yuchen Liu*, Qingya Huang, Li Zhang *Equal contribution
  (Under review) Empirical Methods in Natural Language Processing (**EMNLP 2026**)
  – Challenged the standard attention mechanism of computing value vectors from the residual stream, discovering that deep layers benefit from a context-free value vector that represents original token information, and that models decouple the optimal input for keys vs. values
  – Proposed a new attention variant that directly learns a value vector table for each of the last third of layers, eliminating the V cache entirely and surpassing standard attention on both validation loss and average scores across 21 benchmarks on two model sizes (135M and 780M)

- **High-fidelity Simulations of Human Traits for Testing Agents** 8
  Muyu He, Anand Kumar, Meghana Arakkal Rajeev, Soumyadeep Bakshi, James Zou, Nazneen Rajani
  (Accepted by) Association for Computational Linguistics (**ACL Oral 2026**)
  – Improved a training-free technique for fine-grained, arbitrary control over LLM personality by extracting per-layer activations that correspond to human traits (impatience, confusion, etc.)
  – Developed a robust agentic evaluation framework using activation-controlled human-simulator models, demonstrating effectiveness over LoRA, SFT, and prompting on 1K+ conversations scored by human annotators

- **Evaluating Deductive Reasoning via Detective Games** 5
  Yuan Yuan*, **Muyu He***, Adil Shahid, Jiani Huang, Ziyang Li, Li Zhang *Equal contribution
  (Accepted by) Empirical Methods in Natural Language Processing (**EMNLP 2025**)

- **Commonsense-T2I Challenge: Can Text-to-Image Generation Models Understand Commonsense?** 45
  Xingyu Fu*, **Muyu He***, Yujie Lu, William Yang Wang, Dan Roth *Equal contribution
  (Accepted by) Conference On Language Modeling 2024 (**COLM 2024**)

Work Experience

Research Scientist **July 2025 - Current**
Collinear.ai *Sunnyvale, US*

- *Spider* (popular on-policy distillation framework) ★105
  – Implemented cross-tokenizer alignment to enable on-policy distillation on any teacher-student model pairs, resolving chat-template issues in existing algorithms by HuggingFace
  – Enabled a token-in-token-out protocol for multi-turn rollouts with robust bug patches for standing tokenizers (e.g. Qwen3), extending on-policy distillation to agentic tool use without mismatch issues
- *Computer-use RL environments* (provider for 3 frontier labs on CUA capabilities)
  – Developed a robust automated environment generation pipeline spanning task generation, verifier generation, data seeding, and Docker runtime setup across 11 real-world apps (e.g. Slack, Asana, Figma, GitHub)
  – Discovered systematic insights on fundamental failure modes in CUA of frontier models (Claude, GPT), enabling the design of RL-efficient tasks in the environment with 33–66% Pass@1

Open-source Projects

RiddleHe/nanochat ★ 52

- Built a hackable pretraining stack from karpathy/nanochat that supports easy definition of architecture and rigorous FLOP-controlled ablations
- Shipped a mirrored RL stack for RL algorithm research that allows clean objective definition with production-grade vLLM inference serving

RiddleHe/llm-interp ★ 91

- Authored a series of completely reproducible interpretability scripts to conduct model circuit research
- Produced findings on attention sinks (the existence of massive activations in 1-2 dimensions for Qwen3 models only on the first tokens) and an open-source reproduction of Thinking Machines' finding on LLM decode indeterminism

RiddleHe/gpt-oss-alignment ★ 125

- Jailbroke the RL-aligned GPT-OSS-20B model via simple chat-template removal, revealing the fragility of post-training alignment that relies on priors of the assistant chat format
- Provided a popular technique to arbitrarily manipulate responses from GPT-OSS-20B by steering activations corresponding to traits, and implemented an end-to-end framework for training its sparse autoencoders